

Prototype Shopping List

Steel Beam Cutting Cell - 1:5 desktop prototype | Ravi Mahadeva | 04 October 2026

**TOTAL BUYING COST: about
£1,920**

incl. VAT where known, before postage | with 10% spare for postage and price changes: **about £2,110** | to get moving first (stages 1-2): about £720

Stage	About
Stage 1 - Frame and bed	£340
Stage 2 - Motion and controller	£380
Stage 3 - Arms and tools	£540
Stage 4 - Cameras (vision)	£100
Stage 5 - Safety and enclosure	£410
Stage 6 - Electrical	£110
Consumables	£40
Total	about £1,920

Before you buy

- Prices are approximate (GBP, October 2026) and change - check each one before ordering.
- **Every item has a link** (blue): a product page where it was checked, otherwise that shop's search.
- **Part numbers** (P01 ...) are the same as the balloons in docs/Prototype_Assembly.pdf, the Prototype tab in the app and the CAD - so you can see where each part goes.
- You already have the **Jetson Orin Nano**; it is not in the total.
- Order the aluminium **cut to size** (lengths on the last page) - it saves a lot of work.
- Buy and wire the **safety parts (stage 5) before any motor moves**: the E-stop and safety relay must cut motor power. Have the mains wiring checked by a competent person.
- Cheapest way to start: stages 1 and 2 first (frame and moving gantry, about £720), then arms, cameras and the rest.
- Save on the arms by buying the STS3215 servos and printing the SO-101 parts yourself (about £130 per arm instead of £240). A used Pilz PNOZ s3 is about £75.
- No real cutting on the prototype: the Cutter holds a **pen** that marks the cut lines.

Stage 1 - Frame and bed

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P01	V-slot 20x40 aluminium extrusion	cut to size: 4 x 1500, 6 x 600, 4 x 580 mm (cad/prototype_cut_list.csv) lengths come from the CAD model	11.9 m	£14	£167	Ooznest (cut to size), Amazon UK Open product page
	P02	V-slot 20x20 aluminium extrusion	cut to size: 2 x 1300, 2 x 300, 6 x 100, 1 x 720 mm bed rails and legs; Z axes; camera post	4.5 m	£8	£36	Ooznest Search ooznest.co.uk
	P03	V-slot 20x60 aluminium extrusion	cut to size: 2 x 660 mm the two bridges	2	£16	£32	Ooznest Search ooznest.co.uk
	P04	Corner brackets + M5 T-nuts + M5 bolts	40 cast corner brackets, 200 drop-in T-nuts, M5x8/M5x10 button heads	1 kit	£45	£45	Ooznest, Amazon UK Search ooznest.co.uk
	P05	Rubber feet / levelling feet	M8 levelling feet for the base rails stops it walking on the bench	4	£3	£12	Amazon UK Search amazon.co.uk
	P06	Bed rollers	20 mm aluminium tube x 100 mm on 8 mm silver steel shaft, 2 x 608 bearings each or 3D print the rollers	6	£4	£24	Amazon UK, Simply Bearings Search amazon.co.uk
	P07	Roller brackets	3D printed in PETG (12 off)	12	£0.50	£6	print 3D print / make it
	P08	Outfeed table board	6 mm plywood or MDF 1300 x 180 mm	1	£8	£8	builders' merchant Search amazon.co.uk
	P09	Scrap tray	folded aluminium sheet or 3D printed tray 1300 x 90 mm	1	£10	£10	print / sheet 3D print / make it
Stage total						£340	

Stage 2 - Motion and controller

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P10	V-wheel gantry plate kits	20-series gantry plate + 4 V-wheels (625 bearings) + eccentric spacers 2 per bridge end x 2 bridges + 1 per carriage x 2	6	£18	£108	Ooznest Search ooznest.co.uk
	P11	NEMA 17 stepper motor	1.5-2 A, 45-48 mm body, 5 mm shaft X (one per bridge) and Y (one per carriage)	4	£12	£48	Ooznest, StepperOnline Search omc-stepperonline.com
	P12	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead, 300 mm screw, with brass anti-backlash nut Z axes	2	£25	£50	StepperOnline, Amazon UK Search omc-stepperonline.com
	P13	Z motor bracket (3D printed)	PETG, holds the Z motor on the carriage plate filament counted under Consumables	2	£0	£0	print 3D print / make it
	P14	Lead-screw nut block (3D printed)	PETG block with the brass T8 nut (comes with the motor), fixed to the Z axis filament counted under Consumables	2	£0	£0	print 3D print / make it
	P15	GT2 belt 6 mm	10 m reel (8 m needed: 2 per bridge along the rails, 1 per carriage)	1	£12	£12	Ooznest Search ooznest.co.uk
	P16	GT2 20T pulleys + idlers	4 x 20T 8 mm bore (cross shafts), 2 x 20T 5 mm bore (Y), 8 idlers	1 set	£25	£25	Ooznest Search ooznest.co.uk
	P17	8 mm cross shafts + pillow blocks	2 x 700 mm 8 mm silver steel, 4 x KFL08 bearings, 2 x 5-8 mm couplers one motor drives both sides of each bridge	1 set	£25	£25	Amazon UK, Simply Bearings Search amazon.co.uk
	P18	Home / limit switches	mechanical microswitch with lever, NC wiring one per axis + 2 spare	8	£1.50	£12	Amazon UK Search amazon.co.uk
	P19	FluidNC 6-axis CNC controller (ESP32)	6 driver sockets, USB to the Jetson (e.g. Elecrow / Bart Dring 6x CNC Controller) flash once from the web installer - no programming	1	£40	£40	Elecrow, Tindie Open product page
	P20	TMC2209 stepper driver modules	StepStick format, UART or standalone quiet; one per axis	6	£5	£30	Amazon UK, Ooznest Search amazon.co.uk
	P21	Shielded 4-core motor cable + drag chain	10 m 4 x 0.5 mm2 shielded, 2 m 10x15 drag chain	1 set	£30	£30	Amazon UK, RS Search amazon.co.uk
Stage total						£380	

Stage 3 - Arms and tools

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P22	SO-101 6-servo arm kit (follower)	6 x Feetech STS3215 serial-bus servos + bus servo driver board; print the arm parts in PLA ~£120-150 each if you buy the servos and print everything yourself	2	£240	£480	Unmanned Tech UK (servo kit Pro, unprinted), Seeed Open product page
	P23	Arm mount plate (3D printed)	PETG plate: bolts the arm base under the Z axis filament counted under Consumables	2	£0	£0	print 3D print / make it
	P24	Bus servo adapter	Waveshare Bus Servo Adapter (A) - USB to the Jetson, one per arm often included in the kit	2	£5	£10	Waveshare, The Pi Hut Open product page
	P25	12 V power supply for the servos	Mean Well LRS-100-12 (12 V 8.5 A) - check your servos are the 12 V version	1	£16	£16	RS, Farnell Search rs-online.com
	P26	Cutter 'torch' for the prototype	pen in a sprung 3D-printed holder (marks every cut line on the model beam) no real cutting on the prototype: see docs/PROTOTYPE.md	1	£3	£3	print 3D print / make it
	P27	Handler magnet	12 V 25 mm lifting electromagnet (~2.5 kg hold) + MOSFET module + flyback diode switch it from a FluidNC output	1	£12	£12	Amazon UK Search amazon.co.uk
	P28	Model beams	3D printed UB 305x165 at 1:5 (61 x 33 mm), 1 m long, with 0.5 mm steel strip glued on the top flange the magnet needs steel to grip	4	£4	£16	print 3D print / make it
Stage total						£537	

Stage 4 - Cameras (vision)

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P29	USB wide-angle camera (safety zone)	1080p UVC, 90-120 deg view (e.g. Logitech C920/C922 or an Arducam USB wide-angle) plug and play on the Jetson; YOLO watches the cell	1	£50	£50	Amazon UK, Pimoroni Search amazon.co.uk
	P30	CSI camera (close-up of the Cutter)	Arducam IMX219 for Jetson Orin Nano (22-pin cable), wide-angle lens optional; checks the marked lines	1	£25	£25	Arducam, RobotShop UK Open product page
	P31	Powered USB 3 hub	4-port, own power supply controller + 2 servo adapters + camera	1	£20	£20	Amazon UK Search amazon.co.uk
Stage total						£95	

Stage 5 - Safety and enclosure

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P32	Emergency stop station	yellow box, red 40 mm mushroom, turn to release, 2 NC contacts (e.g. Schneider Harmony XALK178 family) BS EN ISO 13850; one NC to the safety relay channel 1 (and 2) - add an NO/NC block for the Jetson	1	£42	£42	RS, Farnell, TME Search rs-online.com
	P33	Safety relay	Pilz PNOZ s3 24 V DC (dual channel, manual reset) or equivalent cuts the 24 V motor and 12 V servo supplies - this is how a real machine does it	1	£180	£180	RS, Farnell (new) / ~£75 used Search farnell.com
	P34	Door interlock switch	non-contact coded magnetic safety switch, NC (e.g. Pilz PSEN or Schmersal BNS) BS EN ISO 14119; wired into the safety relay	1	£60	£60	RS, Farnell Search rs-online.com
	P35	Reset push button	22 mm blue illuminated, 1 NO outside the enclosure with a view inside	1	£8	£8	RS, Amazon UK Search rs-online.com
	P36	Stack light	24 V LED tower, red / amber / green / blue driven by the Jetson through a relay module	1	£25	£25	Amazon UK Search amazon.co.uk
	P37	4-channel relay module (opto-isolated)	3.3 V logic input, 24 V contacts stack light and magnet	1	£8	£8	Amazon UK Search amazon.co.uk
	P38	IR break-beam sensor pair	5 mm IR break-beam (demonstrates the light-curtain input) NOT a safety light curtain - demo only	1	£6	£6	Pimoroni, The Pi Hut Search thepihut.com
	P39	Polycarbonate sheet 3 mm	front, door and ends (~1.5 m2) - polycarbonate, not acrylic (acrylic shatters)	1.5 m2	£40	£60	plastics stockist Search amazon.co.uk
	P40	Hinges + handle + 20x20 door frame	2 hinges, 1 handle, 2.4 m 20x20	1 set	£25	£25	Ooznest Search ooznest.co.uk
Stage total						£414	

Stage 6 - Electrical

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P41	24 V power supply	Mean Well LRS-150-24 (24 V 6.5 A) - motors, controller, safety circuit	1	£16	£16	RS (0424827) Open product page
	P42	IEC mains inlet with switch and fuse	panel mount, 5 A fuse have the mains side checked by a competent person	1	£8	£8	RS, Farnell Search rs-online.com
	P43	DIN rail + terminals + fuse holders	35 mm DIN rail, 20 terminals, 4 x 5x20 fuse holders, end stops	1 set	£30	£30	RS, Farnell Search rs-online.com
	P44	Control box	ABS or steel enclosure ~300 x 200 x 150 with cable glands holds the Jetson too	1	£30	£30	RS, CPC Search rs-online.com
	P45	Wire + ferrules + crimp tool	0.75 and 1.5 mm ² flexible, ferrule kit	1 set	£30	£30	Amazon UK, CPC Search amazon.co.uk
	P46	Jetson Orin Nano Super (4 GB)	you already have it - docs/JETSON_SPECS.md runs the app, YOLO and the planner	1	£0	£0	- 3D print / make it
Stage total						£114	

Consumables

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P47	PETG and PLA filament	1 kg each brackets, arm parts, model beams	2	£20	£40	Amazon UK, 3DJake Search amazon.co.uk
Stage total						£40	

Aluminium cut list (order cut to size)

OK	Profile	Length (mm)	Quantity	Used for
	20x20 V-slot	100	6	bed leg
	20x20 V-slot	300	2	Z axis
	20x20 V-slot	720	1	camera post
	20x20 V-slot	1300	2	bed rail
	20x40 V-slot	580	4	cross member
	20x40 V-slot	600	6	upright
	20x40 V-slot	1500	2	base rail
	20x40 V-slot	1500	2	top rail
	20x60 V-slot	660	2	bridge

The lengths come from the CAD model (cad/prototype_1to5.step). The build steps, the safety wiring and why each part was chosen are in docs/PROTOTYPE.md.